



ORIENTATION WORKSHOP - 1 JUNE 2026

Hackathon Overview

UN Global Hackathon on AI for Digital Trade Regulatory Analysis

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Know what to BUILD

Know what to SUBMIT

Know how you'll be JUDGED

You're Shortlisted - What Changes Now

Application

Submitted

Round 1

NOW - Build phase

Finale

Bangkok, Oct 2026

In Stage 1, you had to show...

A team profile - Tech Lead + Substantive Lead

A plausible architecture plan - OCR, RAG, modular design

A realistic cost estimate for processing one document

Plans were enough. Not anymore.

In Round 1, you must deliver...

A working engine - Task 1 and Task 2 running end-to-end automatically

Structured output with verbatim citations for every provision found

New evidence your tool discovered beyond the sample kit provided

A plan becomes a product.

Scope of the Competition

AUTOMATION ZONES

Zone 1

Automated Evidence Discovery
Crawl • Retrieve

Zone 2

Intelligent Mapping
Extract • Map • Cite

Both zones required in every submission

Zone 3 Scoring (Optional → extra points)

Read • Assess • Record

RDTII PILLARS

Pillar 6

Cross-border Data Flows
5 indicators • P6-I1 to P6-I5

Pillar 7

Domestic Data Protection
5 indicators • P7-I1 to P7-I5

Both pillars required for all economies

Other Pillars (Optional → extra points)

ECONOMIES

Round 1 — 3 economies (all mandatory)

Singapore • Australia • Malaysia

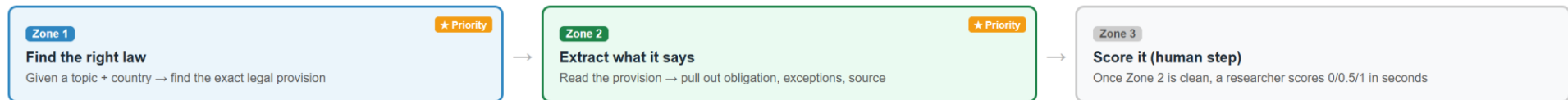
Final Round — 8 additional economies

Thailand • China • India • Indonesia
Russian Federation • Lao PDR
Mongolia • Timor-Leste

Mandatory : 3 out of these additional 8 must be covered

Cover beyond 3+3 (Optional → Extra points)

Automation Zones



Zone 1 — Provision Retrieval ★ Automation Priority

Input: indicator name + country Output: ranked list of relevant legal provisions Hard because: text is unstructured, multilingual, scattered across databases AI tasks: semantic search, dense retrieval, relevance ranking

💡 A Zone 1 tool saves researchers hours of manual database searching.

Zone 2 — Structured Extraction ★ Automation Priority

Input: raw legal provision text Output: structured record (obligation, exceptions, binding flag, source) Hard because: legal language is ambiguous, exceptions are buried AI tasks: NER, relation extraction, text classification, JSON output

💡 A Zone 2 tool gives the researcher everything they need to score quickly — it does not score itself.

Zone 3 — Scoring (lower automation priority)

Input: structured provision record from Zone 2 Output: score 0 / 0.5 / 1 with justification Once Zone 2 output is clean, a researcher can score in seconds Automating scoring without good Zone 2 input just propagates errors

Zone 1 & Zone 2 are the hard automation problems. Zone 3 stays with a human researcher — it is fast once Zone 2 output is clean.

Your Country Assignment

Your tool must work on these specific countries - different challenges in each round.

ROUND 1 - Your current focus

3 countries

Country	Assignment	Language
Singapore (10)	Find & capture amendments and new regulations for Indicators 6 & 7	English
Australia (10)	Find & capture amendments and new regulations for Indicators 6 & 7	English
Malaysia (20)	Find & capture amendments and new regulations (Bonus for a hard challenge)	English

Malaysia is harder - it requires both error-checking AND new data collection.

FINAL ROUND - If you advance

3+ countries

NOT in English, Unorganized websites, Complex legal architecture, No single repository, etc.

Thailand (10)

China (10)

India (10)

Indonesia (10)

Russian Federation (10)

Lao PDR (10)

Mongolia (10)

Timor-Leste (20)

(BONUS for a hard challenge))

Finalists must cover 3 countries in round 1 + at least 3 more countries (round 2) from the list.

How You will be Scored in This Round

40%

Substantive Accuracy

Framework alignment: provisions correctly mapped to RDTII indicators

Discovery of New Evidence: find valid provisions beyond the sample kit

Citation fidelity: exact article and paragraph required; location reference where available

30%

Technical Resilience

Crawling: autonomously navigate live government portals to retrieve source laws

OCR quality: handle scanned/image PDFs; clean, structured, searchable output

End-to-end workflow: crawl -> extract -> map with no manual steps

30%

Architecture

Modular backend: swap LLM or OCR engine without rewriting core pipeline

Audit trail: verbatim snippet in every output row so legal expert can verify instantly

Cost-efficiency: Your actual operational cost

KEY INSIGHT:

1. Finding valid and accurate evidence beyond the provided sample kit is a major differentiator.
2. HTML sources are actually a harder technical challenge than PDFs. A team that correctly handles both PDF and HTML sources is also a differentiator.

Discovery of New Evidence

Both teams receive the same Thailand PDPA sample kit

Team A

Provisions found:

+ Art. 26(2) - cross-border transfer restriction provided

+ Art. 28 - data processor obligations provided

+ Art. 37 - enforcement mechanism provided

No new evidence found

Substantive Accuracy: 20 / 40

VS

Team B

Provisions found:

+ Art. 26(2) - cross-border transfer restriction provided

+ Art. 28 - data processor obligations provided

+ Art. 37 - enforcement mechanism provided

* Art. 16 - consent withdrawal rights

NEW

Substantive Accuracy: More than 20 / 40

Same sample kit. Same 3 known provisions. One team went further - and won the accuracy block.

Citation Fidelity: What Judges See

Same provision - two very different outputs

Field	+ Strong output	- Weak output
Indicator ID (RDTII code)	P6-I1	Cross-border data [not mapping to the RDTII framework]
Article / § (article + page + URL)	Section 26(2), p. 14, Correct URL	Section 26 [Low traceability]
Discovery Tag (NEW / KNOWN)	NEW	-- [missing field]
Verbatim Snippet (verbatim text)	<i>"Personal data controller shall not transfer personal data to a foreign country unless..."</i>	The law restricts transfers of personal data abroad. [paraphrase — not verifiable]
Mapping Rationale (why this indicator)	<i>"S. 26(2) prohibits cross-border transfer as the default. Maps to P6-I1 because restriction is the baseline — controller must justify any transfer."</i>	"This provision is about data transfers." [vague — does not explain the mapping logic]
Verdict	Full credit — auditable	Point deduction — not verifiable

WARNING: Wrong or unclear citation = point deduction. Paraphrased snippets cannot be audited and your points will be deducted.

Technical Resilience & Architecture

19-21 min

Judges look beyond accuracy - they test your engineering decisions.

Live Portal Crawling

10 pts

PASS: Engine navigates laws.go.th hierarchy autonomously, retrieves relevant legal evidence without any manual step

FAIL: Engine only processes pre-downloaded PDFs provided by the team - zero points for crawling

OCR on Scanned PDFs

10 pts

PASS: Scanned gazette pages extracted cleanly, article structure preserved, error below 5%

FAIL: Tool fails or produces distorted text on image-based PDFs - common in older legal documents

Modular Backend

15 pts

PASS: Can swap OpenAI API for Llama 3 (self-hosted) by changing a config value, not rewriting pipelines

FAIL: LLM is hardcoded throughout - replacing it requires rebuilding 60% of the codebase

Audit Trail

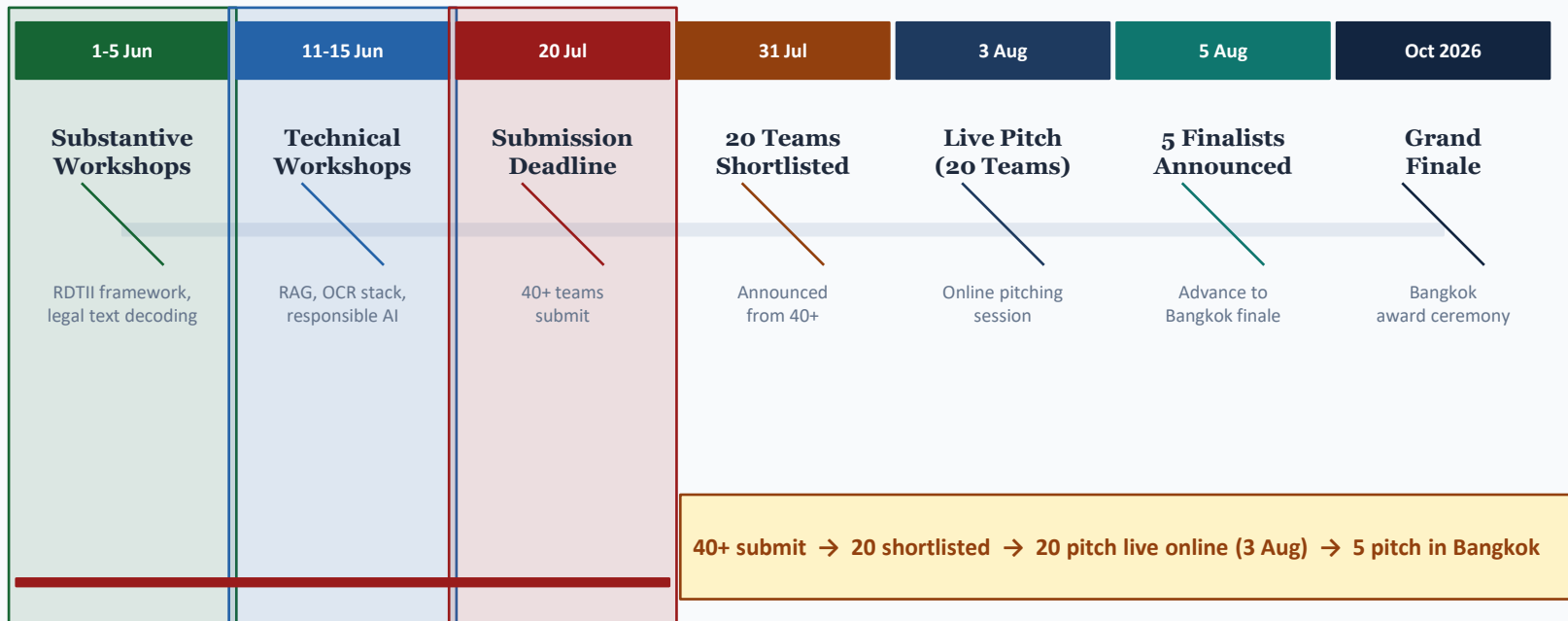
15 pts

PASS: Every output row has a verbatim snippet - legal expert clicks, sees exact source text highlighted

FAIL: Output shows conclusions only - no way to verify the AI did not hallucinate the citation

Timeline: Now to Finale

You have 7 weeks to build a working engine. Here is every key date.



By the end of the series, your Canvas Design should be clear and concrete.

1

Familiar Yourself with the Assignment and the Challenge Framework (RDTII)

The full technical and submission requirements are in the hackathon brief. Both leads should read it - not just the one it seems relevant to.

2

Divisions of Labour in your team

Your tool must prove both Technical Excellence and Substantive Accuracy. Agree now on who owns what before the build window opens.

3

Map your tech stack to Task 1 and Task 2

List the tools you plan to use for: portal crawling, OCR, LLM, citation extraction. Identify any gaps before the technical workshops (11-15 June).

Next: Knowledge Workshop - 4 June, 7:00 PM Bangkok time

What You Must Submit (20 July)

Five deliverables. All required. No exceptions.

#	Deliverable / Key requirement	Watch out for	Format
1	Functional Prototype Task 1 + Task 2, end-to-end. A Quick Start README (env setup + how to run) INSIGHT: A complete README is mandatory for the finale.	Stress test with the reviewer	API / Script
2	Structured Output File	Judges include non-technical reviewers	JSON / CSV
3	Technical Pitch Deck Problem-solution fit; your extraction and mapping logic for Task 1 and Task 2	Judges include non-technical reviewers	Slides
4	Screen-recording Walkthrough Engine processing a scanned/image PDF document, generating correct citations	Max 10 minutes	Video
5	Live Demo + Interview (3 August) Hybrid pitching session - engine must run and produce output in real time	Not a slide presentation	Live

Deadline: 20 July 2026 | 20 shortlists: 31 July | Live pitch: 3 August | 5 finalists: 5 August 2026

Functional Prototype — What "runs" really means

Required behaviour

- A reviewer hands your engine a country + topic.
- Your engine returns structured output (CSV + JSON) with verifiable citations.
- No manual steps in between.
- Works on both text PDFs and scanned/image PDFs.
- Handles inputs you did not anticipate (mis-spelt country, edge cases).
- Switchable open-source fallback if commercial API used.

Quick Start README — mandatory sections

```
# Project name + one-line description
## Setup (3-5 lines)
git clone <repo> && cd repo
pip install -r requirements.txt

## Run (one command)
python run.py --country SG --pillar 6

## Outputs
output.csv, output.json, logs/

## Pinned versions
LLM, OCR, libs – no "latest"

## Open-source fallback (if commercial API)
```

INSIGHT — A complete README is mandatory for the finale. Build it from day one.

Submission Timeline — the four dates that matter

20 JUL

2026



Round 1 submission deadline

All 5 deliverables uploaded

31 JUL

2026



20 shortlists announced

Top 20 advance to live pitch

3 AUG

2026



Live pitch (3 hr)

Hybrid demo + interview

5 AUG

2026



5 finalists announced

Travel grants activated

If you only screenshot one slide, screenshot the timeline. 20 July is your build deadline. 31 July is the rubric gate. 3 August is when your engine has to actually run in front of a panel. 5 August decides who goes to Bangkok.

Structured Output: Recommended Fields

INSIGHT: Design your outputs for users is a differentiator. Judges are both types.

Primary: CSV — opens in Excel, filterable, reviewable row by row by both judge types

Supplementary: JSON — carries richer metadata for technical validation

CSV — Column Structure			Primary output
Column	Purpose	Reviewers/Judges	Req.
Economy	Country being analysed	Both	Yes
Law Name	Full official statute name + year	Policy	Yes
Law Number/Ref	Official act/law number	Policy	Opt.
Last Amended	Year of most recent amendment	Policy	Yes
Indicator ID	RDTII code — e.g. "P6-I1"	Both	Yes
Article / Section	Exact article and paragraph	Policy	Yes
Discovery Tag	"NEW" = independent find; "KNOWN" = sample kit	Both	Yes
Location Ref.	PDF: page no. HTML: URL anchor / path	Both	Opt.
Verbatim Snippet	Exact quoted text — no paraphrasing	Policy	Yes
Mapping Rationale	Why this maps to this indicator (max 300 chars)	Policy	Opt.
Source URL	Direct URL to law on official portal	Both	Yes
Confidence	Model certainty score (0.00–1.00)	Technical	Opt.
Notes	OCR issues, partial doc, bilingual sources	Both	Opt.

JSON — Extra Fields

Carries metadata CSV cannot hold cleanly:

source_pdf_path
Local path to retrieved PDF

ocr_quality_cer
CER score on extraction

processing_time
Seconds to process doc

model_version
LLM + OCR version used

provisions[]
Array — all rows for one law

raw_context
Surrounding text for HITL

Policy judge: filter Discovery Tag = NEW → read Mapping Rationale → verify Verbatim Snippet against Source URL → done.

What a strong row looks like — CSV + JSON

CSV row — Thailand PDPA cross-border (sample)

Economy:	Thailand
Law Name:	Personal Data Protection Act 2019 (PDPA)
Law Number/Ref:	BE 2562 (2019)
Last Amended:	2022
Indicator ID:	P6-I1
Article / §:	Section 26(2)
Discovery Tag:	NEW
Location Ref:	p. 14
Verbatim Snippet:	"Personal data controller shall not transfer personal data to a foreign country unless..." S.26 prohibits cross-border transfer absent adequacy or controller-binding rules – direct evidence for Pillar 6 Indicator 1 (default restriction).
Mapping Rationale:	
Source URL:	https://laws.go.th/pdpa-2562
Confidence:	0.93
Notes:	Original text Thai; English official translation cross-checked.

JSON envelope — same provision, with metadata

```
{
  "economy": "Thailand",
  "law_name": "PDPA 2019",
  "source_pdf_path": "src/pdpa_2019.pdf",
  "ocr_quality_cer": 0.014,
  "processing_time": 47,
  "model_version": "qwen3-32b@2025-09 + tesseract-5.3",
  "provisions": [
    {
      "indicator_id": "P6-I1",
      "article": "Section 26(2)",
      "verbatim": "Personal data controller shall not...",
      "mapping_rationale": "S.26 prohibits cross-border...",
      "discovery_tag": "NEW",
      "raw_context": "Chapter III Sec 26(1)...(3)..."
    }
  ],
  "confidence": 0.93
}
```

CSV serves the policy judge in Excel · JSON serves the technical judge through scripts.

Two reviewer paths — design for both

Policy Judge — reads the law

Workflow inside Excel:

- 1 Open your CSV in Excel.
- 2 Filter Discovery Tag = NEW.
- 3 Read Mapping Rationale (300 chars).
- 4 Click Source URL → verify Verbatim Snippet.
- 5 Score the row. ✓

Technical Judge — reads the code

Workflow inside terminal:

- 1 Clone repo, run Quick Start README.
- 2 Inspect JSON: model_version pinned?
- 3 Inspect ocr_quality_cer values.
- 4 Re-run on hold-out economy.
- 5 Check raw_context for hallucination. ✓

If both loops close in seconds, you score. If they don't, you don't.

What Support You Get

You are not building alone. Here is every resource available to your team.

1 Sample Kit

5-10 legal documents from 1-2 specific countries

Includes machine-unreadable (scanned/image) PDFs

Sample dataset as your testing ground for accuracy

2 5 More Workshops

4 June: Legal structure + end-user perspective (open)

5 June: Hands-on RDTII extraction practice (participants only)

11-15 June: Technical seminars - RAG, OCR, responsible AI, architecture (open + participants)

3 Q&A Portal

Centralized help desk for technical and policy queries

Verified answers from ESCAP and KMITL steering staff

Post a question anytime - guaranteed response before the submission deadline

4 Resource Library

RDTII 2.1 framework documentation and indicator definitions

Sample government portals for testing your crawler

Submission templates: Recommended schema for output, pitch deck, README

Knowledge Portal

Your one-stop-shop to support your build.

- 1 **ESCAP-2025-RDTII-2.1-internal-guide.pdf**
Implicit rules for RDTII data collection and legal research
- 2 **ESCAP-2025-RDTII-2.1-guide.pdf**
Documentation of RDTII structure, methodological notes and descriptions of indicators
- 3 **RDTII 2.1 Round 1 Database.xlsx**
Sample data for Australia, Singapore and Malaysia
- 4 **tool_design_canvas_template.docx**
Template for take-home assignment 2 | Due 4 June
- 5 **Take home assignment 1.docx**
Due 4 June
- 6 **Output_Template.xlsx + PITCHDECK_RegTech.pptx + README.md**
Recommended structure of your outputs and submission
- 7 **Video recordings and presentations from workshops**
Added after each session

And more as we go...

Access the Portal



Scan to request access

What's inside:

RDTII guides
Round 1 database
Assignment templates
Output templates
Workshop recordings

All resources are updated after each workshop session. Check the portal regularly.

Knowledge portal contents (as of today)

